Bio 301 Assignment no 2

Question 1

What are the limitations of Mendelian genetics? Explain.

Answer:

Following are the general limitations of Mendelian genetics:

1) Phenomenal of dominance is heterozygous gene type in all examples in Mendelian genetics but dominance does not occur in every case of contrasting traits.
2) Blending type of inheritance is not described in Mendelian genetics but it may occur in many cases.
3) Sometimes, two alleles co-dominate but this phenomenon is not described in Mendelian genetics.
4) Mendelian genetics does not explain the non-allelic gene interactions such as apistasis, presence of complementary genes etc.
5) It does not explain linkages between different genes.
6) It does not explain the effects of non-disjunction that occur during the formation of gametes.
7) It does not explain the pleioteopic effects.

Question no 2

The gene for plant height in sunflowers has two co-dominant alleles, T1 for tall plants and T2 for short plants. If a tall plant is crossed with a short plant, Draw table and also write Genotype and Phenotype.

Answer:

As plant height show phenomenon of co dominance then the genotype for tall and short plant will be homozygous.

Genotype of tall plant: T1T1

Genotype of short plants: T2T2

Cross between tall and short plant:

	T1	T1
T2	T1T2	T1T2
T2	T1T2	T1T2

Genotype: heterozygous

Phenotype: co dominant.